biomovies manual

The documentation of biomovies, to read offline. The latest version, with the release notes, is at https://biomovies-documentation.readthedocs.io



Introduction



biomovies is a desktop editor of **molecular movies** for **macOS**, **Windows** and **Linux**. Starting from a 3D structure, you build a movie clip by clip, each clip with its own representations and camera, and export it as a video.

What you can do

- **Start from** a **PDB** or **mmCIF** file, a **PDB code**, an **AlphaFold DB** model or a **UniProt** accession, an **MDposit** simulation, or a structure with its **trajectory** (**XTC**, **DCD** or **NetCDF**).
- **Build** the movie in clips: in each one, **representations** (what to show, how and in which colour) and the **camera**; between them, **transitions** and **fades**. Add **descriptions** (text on screen), **labels** and **measurements**, and play the **trajectory** inside the clips.
- **Export** an **MP4 video** (from 720p up to **4K**, also in **3D**), or an **animated GIF**, of the whole movie or of a part of it.
- **Save** the project in a **.biomovies** file, with the structure and the trajectory inside, to edit or share it. **Templates** make the same movie of another structure: start from a ready-made one or save your own.
- **Ask an AI assistant**: the **assistant in biomovies** (a panel in the editor, with a model from Google, Anthropic, OpenRouter or your own computer), or **Claude Code**, **Claude Desktop**, **opencode** or **VS Code's Copilot**, can build and change movies in the open editor, from a request or from a paper.

biomovies is **non-profit experimental** software in its **Beta** version: some **bugs** can still happen. **Save** your projects often and **let us know** when something goes wrong.

Movies made with biomovies are in the **website gallery**, and step-by-step videos in its **tutorials**.

This documentation is also a **PDF to read offline** (in the app too: *Help > Offline Documentation*).

Requirements

	macOS	Windows	Linux
System	macOS 12 (Monterey) or later	Windows 10 or 11	Ubuntu 22.04 or later
Processor	Apple silicon (M1 or later)	64-bit (x64)	64-bit (x64)
Memory	16 GB recommended	16 GB recommended	16 GB recommended

A graphics card helps but is not required: on computers without one (virtual machines, remote desktops) biomovies works, drawing more slowly. An internet connection is needed only to open structures from online databases (PDB, AlphaFold DB, MDposit) and to check for updates.

Installation

See the **installation section**.

Technology

biomovies is built with **Vue** on top of **Electron**, with the **PrimeVue** components. The molecules are drawn by **Mol***.

Acknowledgements

biomovies is built on these libraries, among others (*Help* > *Third-Party Licenses* in the app lists them all):

- **Mol* Viewer: modern web app for 3D visualization and analysis of large biomolecular structures**
- **Mediabunny** (MP4 videos)
- **gifenc** (animated GIFs)
- **Animation Timeline Control**

Credits

Genís Bayarri, Adam Hospital.

Copyright & licensing

This software has been developed by the **MMB group** at the **IRB Barcelona**.

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Installation

Download the file for your computer from the website or from the latest release on GitHub:

System	File
macOS (Apple silicon)	`biomovies-X.Y.Z-arm64.dmg`
Windows (64-bit)	`biomovies-X.Y.Z-x64.exe`
Ubuntu (64-bit)	`biomovies-X.Y.Z-amd64.deb`

biomovies is free software made by a research group, and its installers are not signed with a paid certificate from Apple or Microsoft. So the first time, your computer warns you that it does not know the app. The steps below show how to open it anyway. You do this only once.

macOS

1. Open the **.dmg** file and drag **biomovies** onto the **Applications** folder.
2. Open **Applications** and double-click **biomovies**. macOS says it could not verify that biomovies is free of malware: click **Done**.
3. Open **System Settings > Privacy & Security** and scroll down to **Security**. Next to *"biomovies" was blocked to protect your Mac*, click **Open Anyway**. Confirm with **Open Anyway** and your password or Touch ID.

From then on, biomovies opens with a double-click like any other app.

On **macOS 12 to 14**, steps 2 and 3 are simpler: right-click (or Control-click) **biomovies** in **Applications**, choose **Open**, then **Open** again.

If macOS says that biomovies *is damaged and can't be opened*, the file is fine: it is how macOS sometimes treats apps from the internet that are not notarized by Apple. Open the **Terminal** app, paste this line, press Enter, and open biomovies again:

```
xattr -cr /Applications/biomovies.app
```

Windows

1. If your browser warns about the download, choose **Keep**.
2. Double-click the **.exe** file. Windows shows *Windows protected your PC*: click **More info**, then **Run anyway**.
3. biomovies installs for your user (no administrator password needed) and opens. It leaves a shortcut on the desktop and in the Start menu.

To uninstall it: **Settings > Apps**, **biomovies**, **Uninstall**.

Ubuntu

1. If your browser warns that the file may be dangerous, open the browser's downloads list and choose **Keep** (in Chrome, sometimes under ⋮ as *Download dangerous file*). Browsers warn about any installer that few people have downloaded yet.
2. Double-click the **.deb** file. It opens in **App Center** (**Ubuntu Software** in Ubuntu 22.04): click **Install** and enter your password.
3. biomovies is in your applications: click **Show Applications** and search for *biomovies*.

If the .deb file opens in another program, right-click it and choose **Open With > App Center** (or **Software Install**). If installing that way fails, open the **Terminal** app in the folder of the file and type:

```
sudo apt install ./biomovies-X.Y.Z-amd64.deb
```

with the name of the file you downloaded.

To uninstall it: find **biomovies** in **App Center > Manage** and click **Uninstall**.

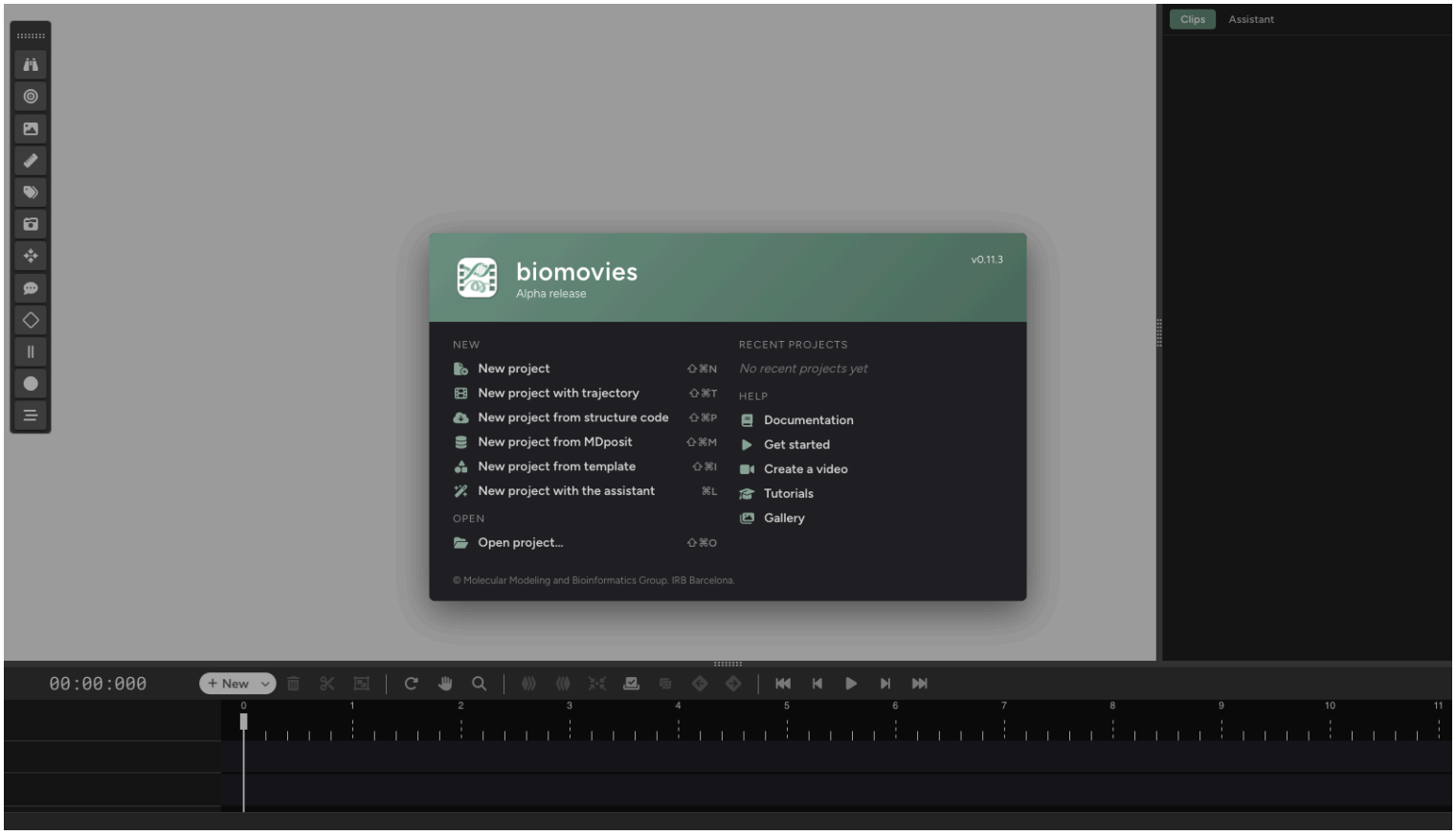
Updates

biomovies tells you when a new version is out, on its start screen and in the editor's status bar. You can also ask: **biomovies > Check for Updates…** in the menu.

- **macOS:** biomovies downloads the new version for you. **Quit and Install** opens it: drag **biomovies** onto **Applications** again and replace the old one.
- **Windows** and **Ubuntu:** biomovies opens the release page. Download the new file and install it as the first time: it replaces the old version.

Your projects are not touched by an update.

Getting started



The start screen

When biomovies opens, the **start screen** offers every way to start a project. Behind it, the editor waits empty: it can be used once a structure is open.

Starting a project

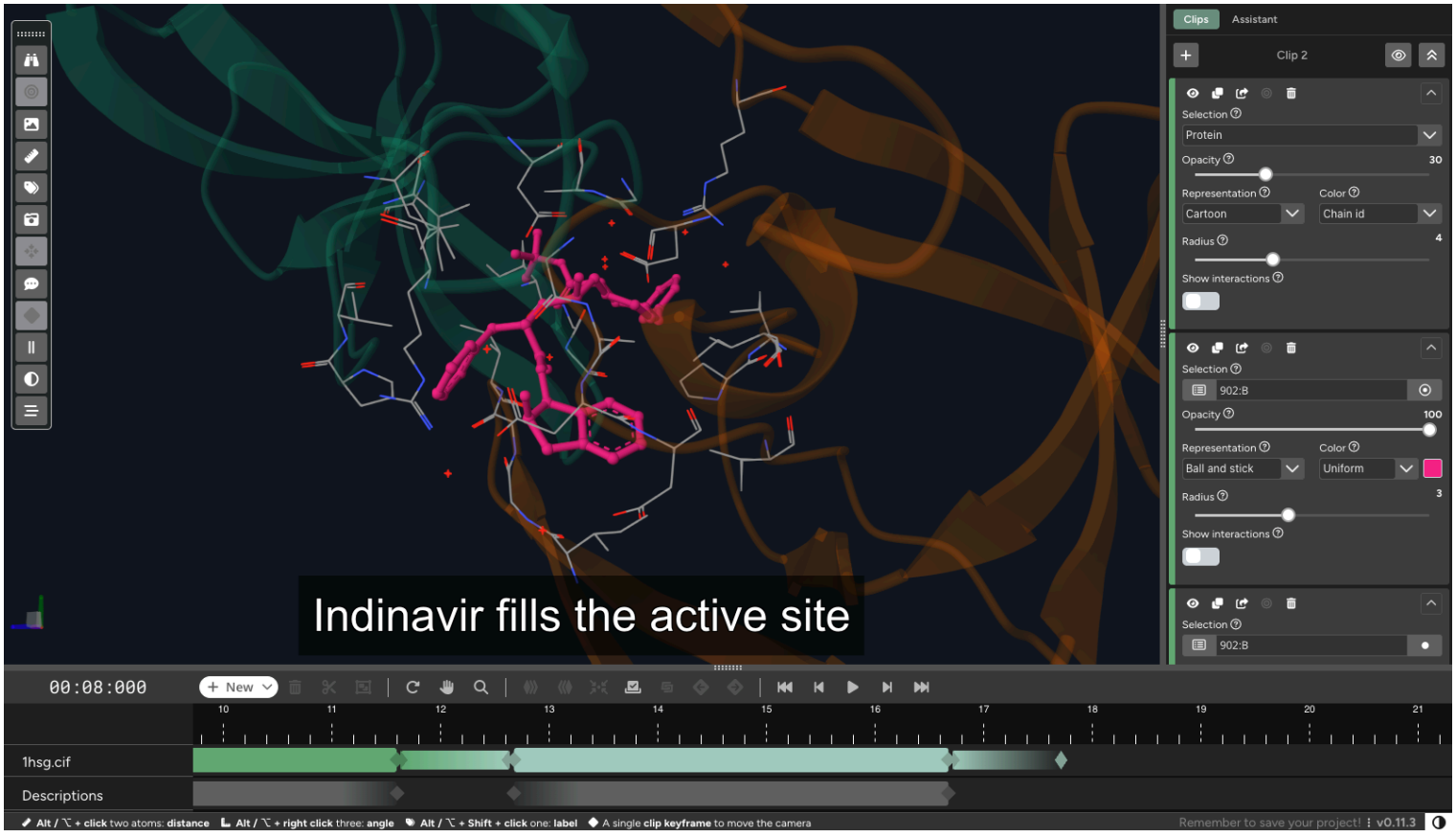
	Shortcut (macOS)	Shortcut (Windows, Linux)
New project: a structure file on your computer (**PDB** or **mmCIF**).	⇧⌘N	Ctrl+Shift+N
New project with trajectory: a structure file (the topology) and its trajectory (**XTC**, **DCD** or **NetCDF**).	⇧⌘T	Ctrl+Shift+T
New project from structure code: a **PDB ID** (from the PDBe), an **AlphaFold DB ID** or a **UniProt** accession (its AlphaFold model), downloaded for you.	⇧⌘P	Ctrl+Shift+P
New project from MDposit: a simulation from the MDposit database, with its trajectory.	⇧⌘M	Ctrl+Shift+M
New project from template: a ready-made movie (or one you saved as a template) applied to your structure.	⇧⌘I	Ctrl+Alt+I
New project with the assistant: describe the movie you want; the assistant opens the structure and builds it.	⌘L	Ctrl+L
Open project...: a **.biomovies** file.	⇧⌘O	Ctrl+Shift+O

The same entries are in the **Project** menu. On the right, **Recent projects** opens the last projects with one click, and **Help** links to this documentation, the tutorials and the gallery. When a new version of biomovies is out, the start screen says so.

A few things to know:

- **From a structure code**, a PDB entry opens **as deposited**. Tick **Biological assembly** to open the assembly instead (for example the dimer of 6LU7, where the file holds one copy).
- **With a trajectory**, choose both files at once in the same dialog (keep them in the same folder). The trajectory must belong to that topology: check it first in a viewer such as VMD. A big trajectory takes a while to load, and makes the **.biomovies** file big: the project file keeps the structure and the trajectory inside. To share only the movie's settings, export it as a **template** (*Project > Export Project As Template*): a small file, without the structure.
- **From a template**, a window shows the ready-made templates, with a preview of each, and lets you choose your own **.tbiomovies** file. Then the start screen asks for the structure to use with it.

The editor



The editor: the stage with the tools strip, the right panel, the timeline and the status bar

- **Stage**, in the middle: the structure as it looks at the timeline cursor. Drag to turn it, scroll to zoom. The **tools strip** on its left has the tools for the stage (background, measurements, labels, the camera…).
- **Right panel**, with two tabs. **Clips**: what is under the timeline cursor (for a clip, its **representations**, one card each; for a description, its text and style). **Assistant**: the assistant in biomovies, to ask for a movie or a change.

- **Timeline**, at the bottom: the movie in time. The **+ New** button adds clips, transitions, fades and descriptions; the buttons next to it edit the timeline and play the movie. Below, a row for the structure's **clips** and **transitions**, and one for the **descriptions**.
- **Status bar**, at the very bottom: tips about what the mouse is over, whether the project is saved, the version.

Your first movie

A short movie of the HIV protease with its inhibitor (PDB **1HSG**): the whole protein turning, then a close-up of the inhibitor. About ten minutes.

1. Open the structure. On the start screen, click **New project from structure code**, type `1HSG` and click **Open**. The editor opens with **one clip of 1 second** that shows every atom in ball and stick.

2. Make the clip longer. A clip starts and ends with a **keyframe** (the small diamonds in the timeline). Drag the clip's last keyframe to the right, to about 5 seconds.

3. Choose what it shows. With the timeline cursor on the clip, the right panel shows its representation. Set **Selection** to *Protein*, **Representation** to *Cartoon* and **Color** to *Chain id*. Then click **+** at the top of the panel to add a second representation: *Ligand*, in *Ball and stick*, in a colour that stands out.

4. Set the camera. Click the clip's **first keyframe** and turn and zoom the structure on the stage: this is the view the clip starts with. Click its **last keyframe** and turn the structure a quarter of a turn: while the clip plays, the camera goes from the first view to the last. Press **Shift+P** (or the play button) to watch it.

5. Add a second clip. Double-click the clip's bar to select it (or ⌘+click, Ctrl+click), then **+ New > Cloned Clip After**: a copy with the same representations, whose camera starts where the first clip ends. Make it 4 seconds long, as in step 2.

6. Make it a close-up. On the second clip, set the camera at its keyframes to a close view of the inhibitor (step 4). On the protein's card, lower the **Opacity**, so that the inhibitor stands out and the protein stays as context.

7. Join them with a transition. A transition needs a little room: double-click the second clip to select it and drag it a bit to the right, leaving at least 0.4 seconds between the clips. Click the **last keyframe** of the first clip, **Shift+click** the **first keyframe** of the second, and choose **+ New > Transition**. During the transition, the camera flies from one clip to the other and the representations fade from one to the other.

8. Add a text. Choose **+ New > Description**: the first description goes at the start of the movie, 1 second long (drag its last keyframe to make it last as long as the first clip). Type its text in the right panel (for example *HIV protease*) and choose where it goes on the screen.

9. End with a fade. Click the last keyframe of the second clip and choose **+ New > Fade Out**.

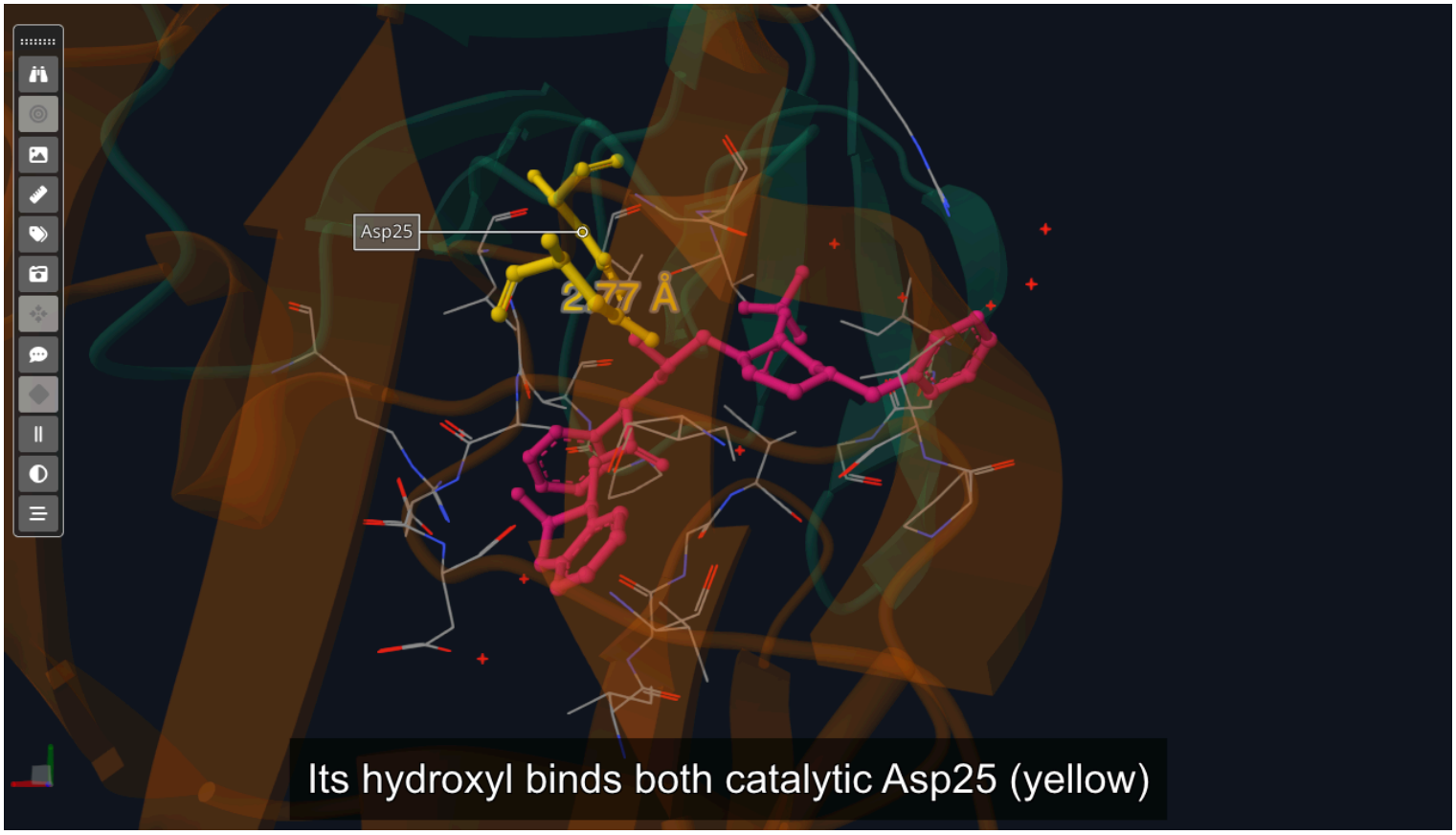
10. Save and export. Save the project with **Project > Save Project** (⌘S on macOS, Ctrl+S on Windows and Linux). Then **Create Video > New Movie**: choose the size (720p to 4K) and click **Create video**. When it is done, watch it in the window and click **Save…**.

Every step can be undone with **Edit > Undo** (⌘Z, Ctrl+Z).

What next

- The editor, area by area: the stage and its tools, the timeline, representations, trajectories and descriptions.
- **Exporting videos**: sizes up to 4K, 3D videos, animated GIFs, a part of the movie.
- **AI assistants**: ask the assistant in biomovies, or Claude Code, Claude Desktop, opencode or VS Code's Copilot to build a movie like this one for you.
- The **tutorials** and the **gallery** on the website.

The stage



The stage, with a label (Asp25) and a distance

The **stage** shows the structure as it looks at the timeline cursor. What the mouse does on it depends on what is selected in the timeline, and the status bar says it at each moment:

- **One clip keyframe selected**: the mouse moves the **camera**, and the view is saved into that keyframe. This is how a clip's camera is set: the view at its first keyframe, the view at its last, and the camera goes from one to the other while the clip plays.
- **The timeline cursor on a clip, nothing selected**: the camera stays where the timeline puts it. Hovering shows what is under the mouse, and the Alt clicks below add labels and measurements to that clip.
- **Anything else** (the movie playing, the cursor on a transition, several keyframes selected): the stage only shows.

To look around without changing any keyframe, use **Explore**.

Mouse and trackpad

Action	What it does
Drag	Turns the structure.
Right-drag	Moves the structure.
Scroll, or pinch on a trackpad	Zooms in and out.
Hover an atom	Lights it, its letter in the sequences, and its name in the legend.
Alt+click two atoms	A **distance** between them, in ångströms.
Alt+right-click three atoms	An **angle** between them, in degrees.
Alt+**Shift**+click an atom	A **label** with the name of its residue (*ASP 25 A*); change it in the *Labels* tool.

On macOS, Alt is the ⌥ **Option** key. Click on the atoms themselves: a click on a bond or on nothing does not count. Labels, distances and angles belong to the clip under the cursor: they are shown while it is on screen, and they fade with it in its transitions.

The tools strip



The tools strip

The strip on the left of the stage. Drag it by its handle (the dotted bar at its top) to put it anywhere on the stage, and double-click the handle to fold it. *View* > *Tools* hides it.

Tool	What it does
Explore	Look around freely, without changing the keyframes (see Exploring the stage). With the timeline stopped and its cursor on a clip.
Center	Brings the whole structure into view, saved into the selected keyframe.
Background	The colour behind the structure, for the whole movie.
Measurements	The distances and angles of the clip: their size, their colour, removing them. A label size of 0 leaves the line without its value.
Labels	The labels of the clip: their text, size, colour and background (the switch turns the background off), and their style: *Placed by itself* (set apart from the atom with a line, the text keeping just outside the structure as it turns; drag it to choose the side and how far), *Where I drag it* (the text stays where you drop it), or *3D text at the atom* (no line).
Camera	Perspective or orthographic projection, for the whole movie.
Center view in atom	Then click an atom: the camera turns to centre it, saved into the selected keyframe.
Legend	Shows, at the bottom right of the stage, the model, chain, residue and atom under the mouse.
Keyframe editor	The selected keyframe's camera in numbers: a rotation around each axis (in degrees) and a translation along each axis (in ångströms). For exact moves, such as a half turn.
Video frame	Shows which sides of the stage a video leaves out, so that nothing important falls outside it. Turn it on before setting the cameras of a movie you will export.
Ambient occlusion	Contact shading: soft shadows where atoms are close, which make the shapes easier to read. For the whole movie.
Sequences	Shows the chains' sequences under the stage (see The sequences).

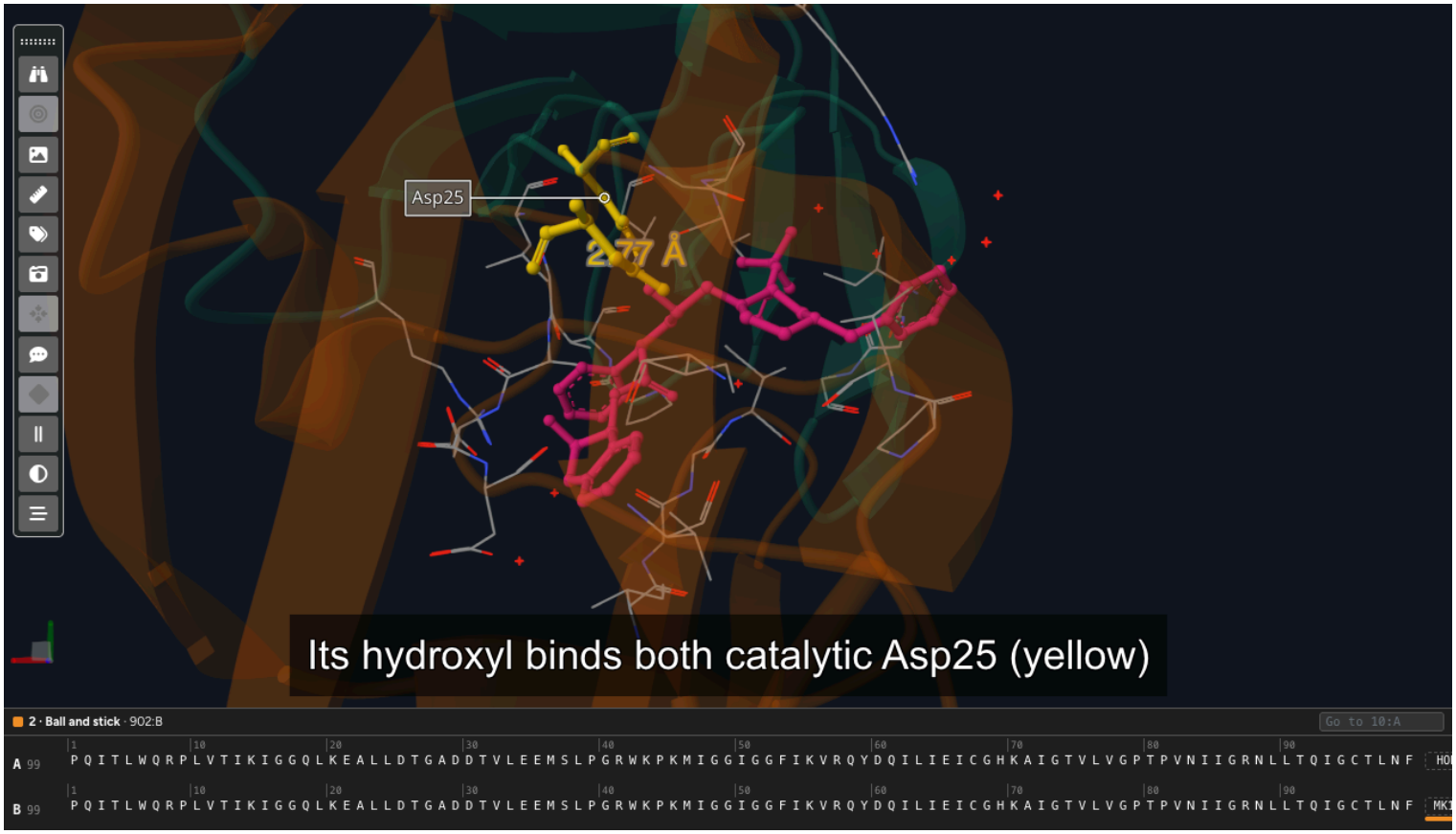
Center, **Center view in atom** and the **Keyframe editor** work on one selected clip keyframe (Center and Center view in atom also while exploring).

Exploring the stage

With the timeline stopped and its cursor on a clip, **Explore** (or *View > Explore Stage*) frees the stage from the timeline: turn, zoom, centre and hover as you like, and no keyframe changes. The timeline waits under a notice meanwhile. Labels, distances and changes to the representations are still saved into the clip under the cursor.

- **Back to timeline** (or **Esc**, or the Explore button again) puts the timeline's camera back.
- **Use this view**, with the timeline cursor on one of the clip's keyframes, copies the camera you found into that keyframe and goes back to the timeline.

The sequences



The sequences under the stage

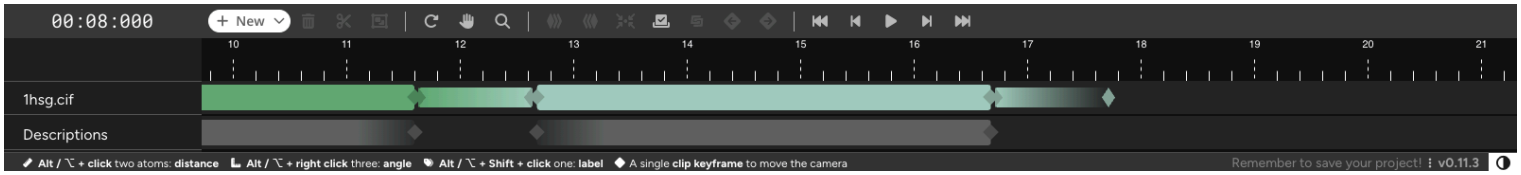
The **Sequences** tool shows a strip under the stage with every chain: its name, its number of residues and its letters, numbered every 10, followed by its ligands and ions as named blocks and its waters as one block. Two chains fit at once; scroll for the rest, or type a residue number in the box at its top and press Enter to go to it.

- Hovering a letter lights its residue on the stage, and hovering an atom on the stage lights its letter.
- Under each chain, a lane shows the residues of the **representation selected** in the right panel (lighter where it draws only part of a residue).
- The strip also **edits** that representation: a click adds a residue or takes it out, a drag adds a range, and Shift+click extends from the last residue clicked. A ligand's block counts as a residue, the waters block as the chain's waters. The representation's selection becomes the list of residues it draws.

Arranging the editor

- **Sizes**: drag the dotted bars between the stage and the right panel, and between the stage and the timeline.
- **View** menu: shows or hides the **Stage**, the **Timeline**, the **Representations** panel, the **Tools** strip and the **Status Bar**.
- **Light theme**: *View* > *Light Theme*, or the round button at the right end of the status bar.
- The **status bar** at the bottom says what the mouse can do where it is, whether the project has unsaved changes, and the version.

The timeline



The timeline: three clips with transitions and a fade out, and their descriptions

The **timeline**, at the bottom of the editor, is the movie in time. It has two rows:

- the **structure** (the row named after its file): **clips**, and the **transitions** and **fades** between them;
- **Descriptions**: the texts over the movie (see Descriptions).

Every item starts and ends with a **keyframe**, the small diamond at each end of its bar. The **cursor** (the white line) marks the time the stage shows, and the **timer** on the left says it in minutes, seconds and milliseconds. Times go in steps of 0.04 s, 25 frames per second, and items keep at least one step apart.

What each item is

Item	What it does
Clip	Shows its own **representations**, **labels** and **measurements**, under a **camera** that goes from the view at its first keyframe to the view at its last while it plays.
Transition	Between two clips: the camera flies from where the first ends to where the second starts, and the first clip's representations fade into the second's.
Fade in, fade out	Before a clip, from an empty stage; after it, to an empty stage.
Description	A text over the movie, with its own times, effects to appear and disappear, and a place and size that can change along its time.

Moving around

- **Click** the timeline to move the cursor there, or drag the cursor.
- **Play** the movie with the player buttons on the right of the toolbar, or with the keyboard: **Shift+P** plays and pauses, **Shift+Left** and **Shift+Right** go one frame back or forward (hold the buttons to keep going), and **Shift+Up** and **Shift+Down** go to the first and last frame.
- **Zoom** the timeline with Ctrl+scroll or a trackpad pinch, or with the **Zoom** button (click to zoom in, ⌘+click or Ctrl+click to zoom out). The **Pan** button lets you drag the whole timeline sideways. **Reset timeline** puts the zoom back, leaves pan mode and takes the cursor to the start.

Selecting

Gesture	Selects
Click a keyframe	That keyframe. A single clip keyframe is what the stage's camera edits.
Double-click a bar, or ⌘+click / Ctrl+click it	The whole item (clip, transition, fade or description).
Shift+click	Adds a keyframe or a bar to the selection, or takes it out.
Drag a rectangle over an empty part	Every keyframe inside it.
Select all keyframes button	Every keyframe of the timeline.

Moving and resizing

- **Drag a keyframe** to make its item longer or shorter. A keyframe cannot go past the next one, so to make room, first move what follows.
- **Drag a selection** to move it: a whole item, or everything selected at once. To make room after a clip, drag a rectangle over everything after it and drag it to the right.
- **Move left** and **Move right** in the toolbar move the selected keyframes one step (0.04 s); hold them down to keep going.

Adding items

The **+ New** button (and *Timeline* > *New Item* in the menu):

Entry	Needs	What it adds
Clip After, **Clip Before**	A selected clip (or one of its keyframes).	A new clip, every atom in ball and stick, whose camera starts where the selected clip ends (or, before it, ends where it starts). *After* goes right after it, or at the end of the timeline when there is no room; *Before* only when there is room.
Cloned Clip After, **Cloned Clip Before**	A selected clip.	The same, but with a copy of the clip's representations, labels and measurements: the usual way to go on with a movie.
Transition	The last keyframe of a clip and the first of the next one, selected (click, then Shift+click), at least 0.4 s apart.	A transition filling the gap between them.
Fade In	A clip's first keyframe selected, with room before it (about 0.5 s).	A fade in before the clip.
Fade Out	A clip's last keyframe selected, with room after it (about 0.5 s).	A fade out after the clip.
Description	For the second one on, a selected description.	The first description goes at the start of the movie, 1 s long. Then *Description After* and *Description Before* add one next to the selected description, and the *Cloned* versions copy its text and style. See Descriptions.

When something is missing (no clip selected, not enough room), a message says what to do.

The toolbar



The toolbar

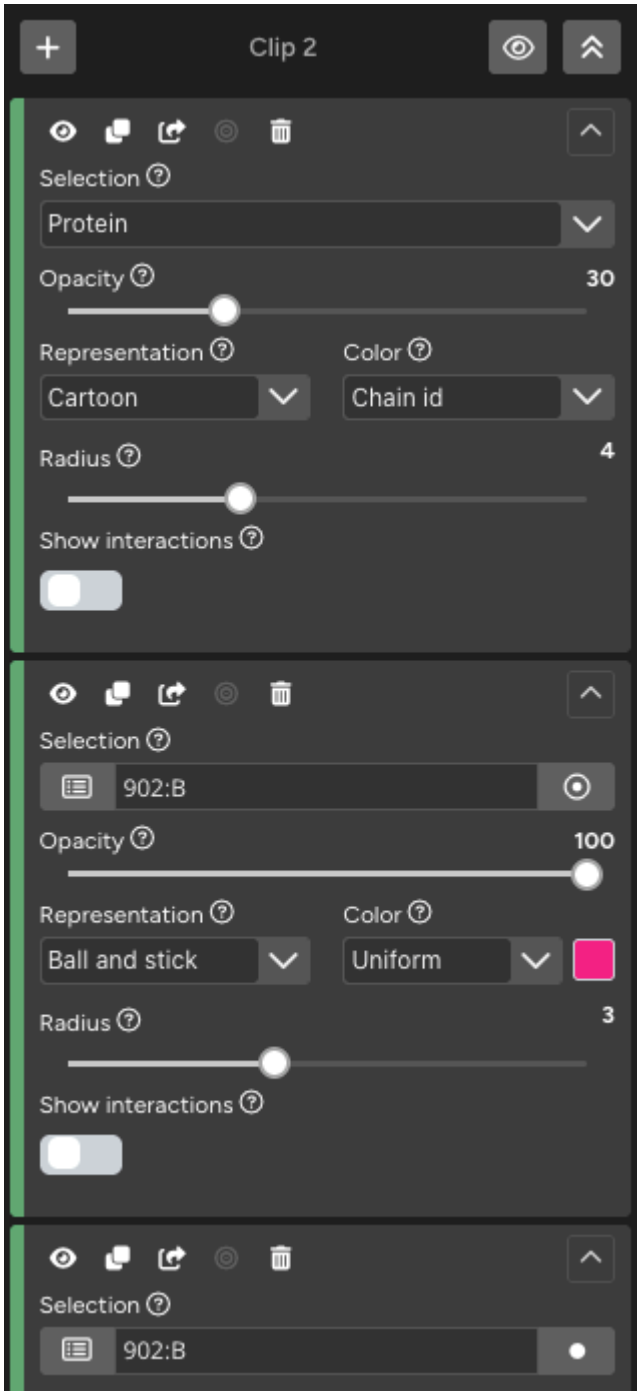
From left to right, after **+ New**:

Button	What it does
Remove	Removes the selected clip, transition, fade or description. Click it twice, so that a single click does not remove anything by mistake.
Split	Cuts the selected clip or description in two at the cursor. Double-click it (or ⌘/Ctrl+click it) where it must be cut: the cursor goes there. Nothing changes in what plays: the second part goes on with the same camera, trajectory and representations.
Join	Joins two consecutive clips, or descriptions, into one: select the last keyframe of the first, Shift+click the first keyframe of the second.
Reset timeline	Zoom back to the start, pan mode off, the cursor at the beginning.
Pan	Drag the whole timeline sideways.
Zoom	Click to zoom in, ⌘/Ctrl+click to zoom out.
Align with the first, **Align with the last**	With several keyframes selected: gives them all the camera of the first (or the last) of the selection.
Align with the cursor keyframe	Gives the selected keyframes the camera of the one under the cursor (click it first, then Shift+click the others).
Select all keyframes	Every keyframe of the timeline.
Associate transition to clip	Links a transition to a clip, for example after replacing the clip it went with. Select the transition's first keyframe and the last keyframe of the clip before it, or the transition's last keyframe and the first keyframe of the clip after it.
Move left, **Move right**	Move the selected keyframes one step.
Player	First frame, previous frame, play / pause, next frame, last frame.

With a trajectory, the right end of the toolbar also shows the trajectory frame at the cursor (see Trajectories).

Every change can be undone with **Edit > Undo** (⌘Z, Ctrl+Z) and redone with **Edit > Redo**. All the shortcuts of the *Timeline* menu are in Tips and shortcuts.

Representations



The right panel: the representations of clip 2

A **representation** is a set of atoms drawn in a style and a colour: the protein as a cartoon, a ligand in ball and stick, the water as small spheres. **Each clip has its own**: a movie changes what it shows from clip to clip, and a transition fades one clip's representations into the next one's.

The **right panel**'s **Clips** tab shows the representations of the clip under the timeline cursor (with the timeline stopped and nothing selected) or of the clip whose keyframe is selected, one card each, with the clip's name at the top. Changes are saved into that clip at once, and the transitions and fades next to it follow them.

At the top of the panel:

Button	What it does
+	A new representation in this clip: every atom, in ball and stick.
Paste	Appears after *Copy to other clips* on a card: adds that representation to this clip.
Hide all / **Show all**	Hides or shows every representation of the clip.
Collapse all / **Expand all**	Folds or unfolds every card.

A representation's card

The buttons at the top of each card:

Button	What it does
Hide / **Show**	Hides the representation in this clip without removing it.
Clone	A copy of the representation in the same clip, to change from there.
Copy to other clips	Copies it, to paste it into other clips (see Copying to other clips).
Center in this representation	Points the selected keyframe's camera at it.
Remove	Removes it from the clip.
Collapse / **Expand**	Folds or unfolds the card.

Below them: **Selection** (which atoms), **Opacity**, **Representation** (the style), **Color**, **Radius** (for some styles) and **Show interactions**.

Selection

Which atoms the representation draws. The list offers the usual sets:

Group	Entries
Everything	*All*
Kinds of molecule	*Protein*, *Nucleic*, *RNA*, *DNA*, *Polymer*, *Hetero*, *Ligand*, *Ion*, *Lipid* (the membrane lipids: POPC, DPPC, cholesterol…), *Saccharide / Sugar*, *Water*, *Hydrogen*
Parts of the protein	*Backbone*, *Side Chain*, *Side Chain Attached* (the side chains with the backbone trace they hang from)
Secondary structure	*Helix*, *Sheet*, *Turn*
Kinds of residue	*Small*, *Nucleophilic*, *Hydrophobic*, *Aromatic*, *Amid*, *Acidic*, *Basic*, *Charged*, *Polar*, *Non Polar*, *Cyclic*, *Aliphatic*
Others	*Bonded*, *Ring*

Custom selections

Choose **Custom** in the list to type a selection; the button on the left of the box goes back to the list. Residue numbers and chain names are the file's own.

Type	To select
`145`	Residue 145 (in every chain).
`-5`	A negative residue number.
`52A:H`	Residue 52 with insertion code A, in chain H.
`10-20`	Residues 10 to 20 (`-7--6` with negative numbers).
`:A`	Chain A.
`10-20:A`	Residues 10 to 20 of chain A.
`145:A.SG`	One atom: SG of residue 145 in chain A.
`ligand` , `water:A`	A keyword, in every chain or in one: `*` (all), `protein` , `nucleic` , `polymer` , `ligand` , `ion` , `lipid` , `sugar` , `water` , `backbone` , `sidechain` , `sidechainAttached` , `helix` , `sheet` , `ring` .
`ligand, 25:A`	Several of the above, separated by commas.
`not ion, protein`	Everything except what follows `not` .

When a selection matches no atoms, biomovies says so, naming what was asked, and leaves the representation as it was.

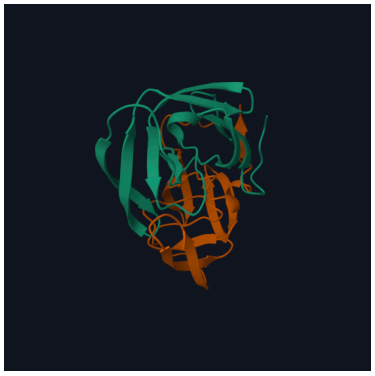
Atoms around a selection

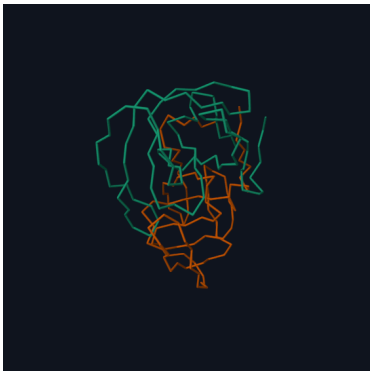
The round button on the right of the custom box turns the selection into the atoms **within a distance** of it: for example, `902:B` with a distance of 5 Å gives the pocket around a ligand. **Distance** sets how far (1 to 50 Å), and **Whole residues** takes every residue with at least one atom in reach, whole, instead of only those atoms.

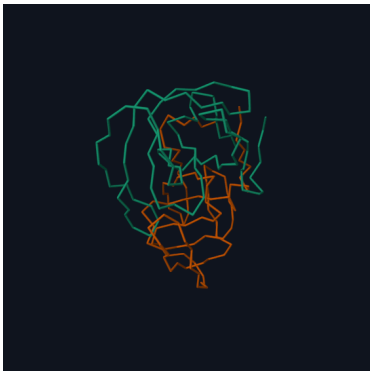
Opacity

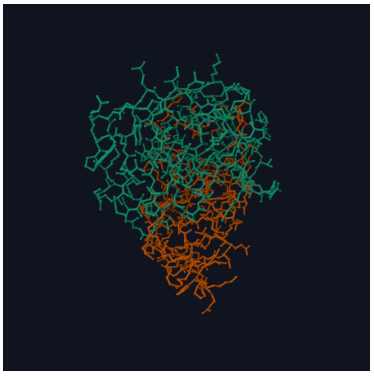
From 0 (invisible) to 100 (solid). Fading the context lets the subject of a clip stand out: for example, the protein at 20 to 40 while the ligand stays solid.

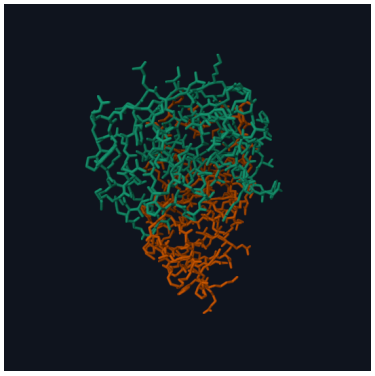
Representation

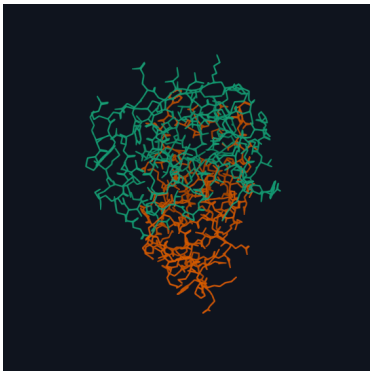

Cartoon

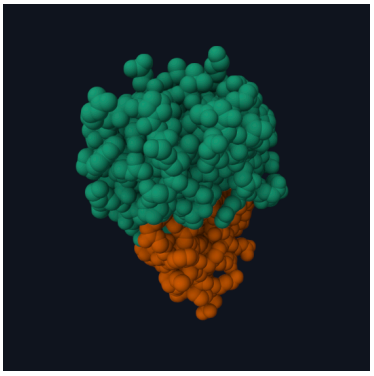

Backbone

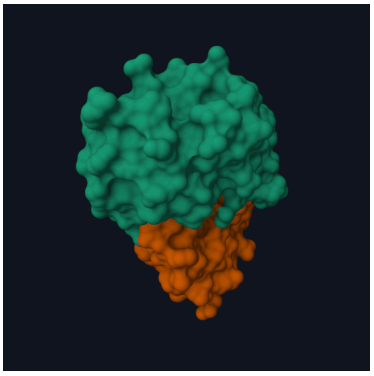

Tube


Ball and stick


Licorice


Line


Spacefill


Surface

The style:

Style	Draws
Cartoon	Helices as spirals, sheets as arrows, loops as tubes: the usual view of a protein's fold.
Ribbon, **Rope**, **Tube**, **Trace**, **Backbone**	Simpler drawings of the chain's path.
Ball and stick	Atoms as balls, bonds as sticks.
Licorice	Bonds as smooth tubes with round joints, every atom the same size.
Hyperball	Atoms and bonds melting into each other.
Line	Bonds as thin lines: light, good for the surroundings of a ligand.
Spacefill	Every atom as a sphere of its own size: the volume a molecule fills.
Surface	The molecular surface.
Base	The bases of nucleic acids.

Radius sets how thick the drawing is, for the styles that have one (Cartoon, Backbone, Ball and stick, Licorice, Spacefill, Base).

Color

The colour scheme:

Scheme	Colours by
Uniform	One colour, chosen with the colour box next to the list: the usual way to make the subject of a clip stand out.
Element	The kind of atom (carbon grey, oxygen red, nitrogen blue...).
Chain id, **Chain index**	Chain.
Secondary structure	Helix, sheet, loop.
Residue index	Position along the chain: a rainbow from one end to the other.
Residue name	Kind of residue.
Hydrophobicity	How hydrophobic each residue is.
B-factor	The B-factor column of the file.
pLDDT (confidence)	A predicted structure's confidence, in AlphaFold's colours (see below).
Atom index, **Model index**, **Random**	Atom order, model, or at random.

pLDDT (confidence) colours a predicted structure (AlphaFold, ColabFold, ESMFold) the way AlphaFold DB does: very high confidence dark blue, confident light blue, low yellow, very low orange. It reads the file's own confidence scores, or the B-factor column, where predictions store pLDDT. For an experimental structure (whose B-factors are not confidence scores) or a file without scores, it is greyed out in the list, with the reason under its name.

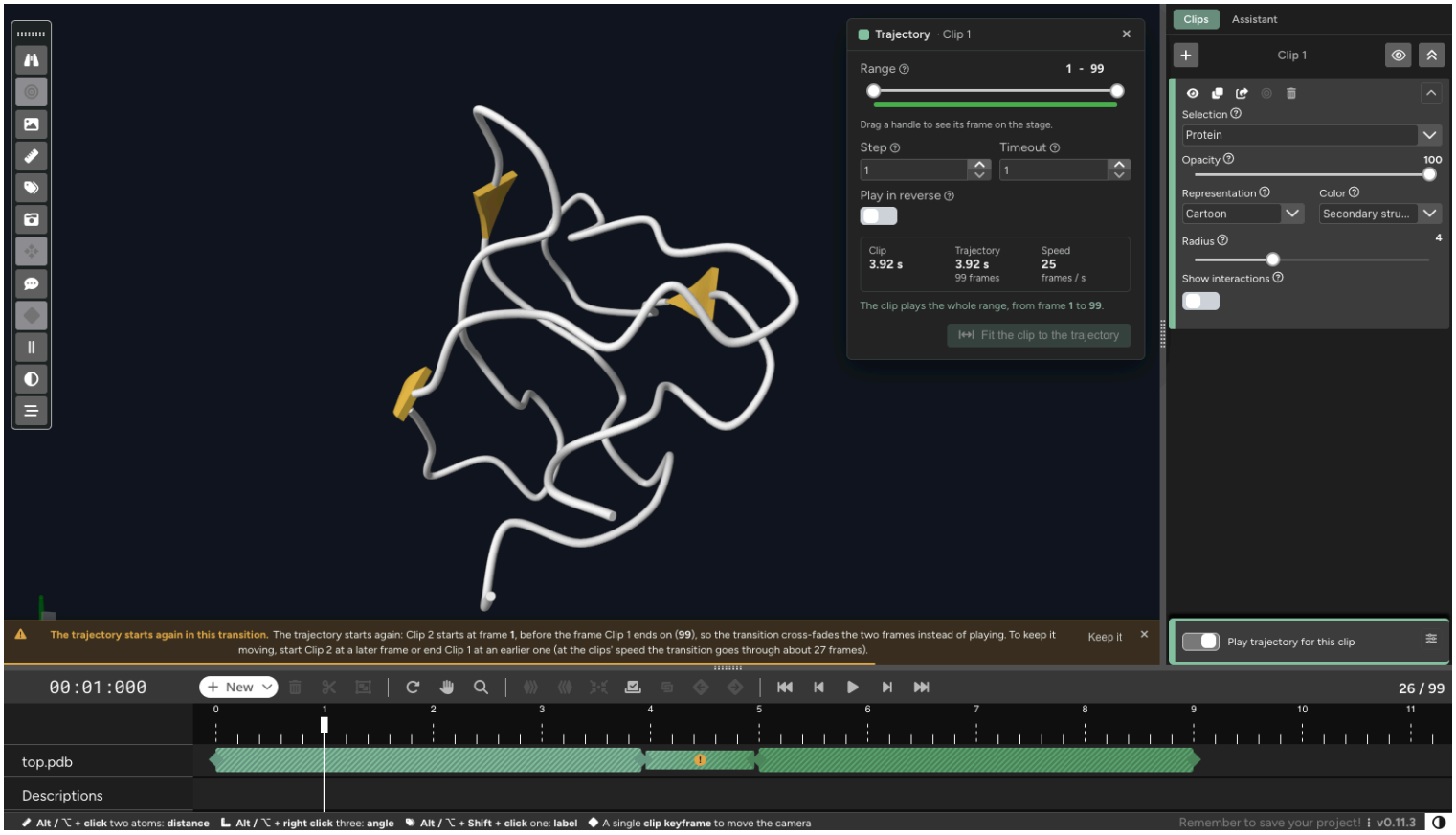
Show interactions

Draws the non-covalent interactions of the selection as dashed lines: hydrogen bonds, ionic, hydrophobic, π-stacking, halogen bonds and metal coordination. The atoms involved are drawn in ball and stick, whatever the representation's style. For example, a ligand with *Show interactions* on shows how it holds in its pocket.

Copying to other clips

Copy to other clips on a card copies the representation. Then, in any other clip, the **Paste** button next to **+** adds it, as many times as wanted. A clip that already has the same representation does not get it again, and a message says so. The copy is kept until the project is closed.

Trajectories



The trajectory window, next to the right panel

A project can have one **trajectory** of its structure: a molecular dynamics simulation, for example. Each clip decides whether it **plays** it, and which frames, how fast and in which direction. A clip that does not play it shows the first frame of its range, still.

Adding a trajectory

- **New project with trajectory** on the start screen: the structure (the topology) and its trajectory (**XTC**, **DCD** or **NetCDF**), chosen together.
- **New project from MDposit**: a simulation of the MDposit database comes with its trajectory.
- **Project > Load Trajectory…**: adds a trajectory to the open project, if it has none yet.
- **Project > Remove Trajectory…** takes it out.

The trajectory must belong to the structure: same atoms, in the same order. A big trajectory takes a while to load, and it is saved inside the **.biomovies** file, which grows with it.

Playing it in a clip

At the bottom of the right panel, **Play trajectory for this clip** turns it on for the clip under the cursor. The clip gets stripes in the timeline, and the **trajectory window** opens with its settings. The button next to the switch opens the window again.

The window floats over the editor: the timeline and the stage stay usable, and it follows the clip the panel shows. Move it by its title; **Esc** or its cross closes it.

Setting	What it does
Range	The first and last frames the clip plays. Type them or drag the slider's handles: the stage shows the frame of the handle you drag. With the keyboard: ⌘+⌥+1 / 2 (Ctrl+Alt+1 / 2) move the first frame down or up, ⌘+⌥+3 / 4 (Ctrl+Alt+3 / 4) the last one.
Step	How many frames it moves at each playing step: 1 plays every frame, 2 every other frame...
Timeout	How many timeline steps (0.04 s each) between playing steps: 1 is the fastest.
Play in reverse	Plays the range from its last frame to its first.

Below them, the window adds it up: how long the **clip** lasts, how long the **trajectory** needs with these settings (and how many frames it plays), and the **speed** in frames per second. A line says how they match:

- the clip plays the **whole range**;
- the range ends **before the clip** does, and the last frame stays on screen for the rest of the clip;
- the clip is **too short**, and ends before the last frame of the range.

The bar under the slider shows the same over the frames. **Fit the clip to the trajectory** makes the clip exactly as long as its range needs. When the next item is in the way, biomovies asks first, then moves everything after the clip later to make room (a description that would be cut too short is removed, and the question names it).

With a trajectory, the right end of the timeline's toolbar shows which frame the cursor is on, among the frames the clip plays (the trajectory's own frame number in its tooltip).

Through transitions

When the clips on both sides of a transition play the trajectory, it keeps playing through the transition: both sides show the same frame, going from the frame the first clip ends on to the frame the second starts on. With the cursor on the transition, the trajectory window says how that goes:

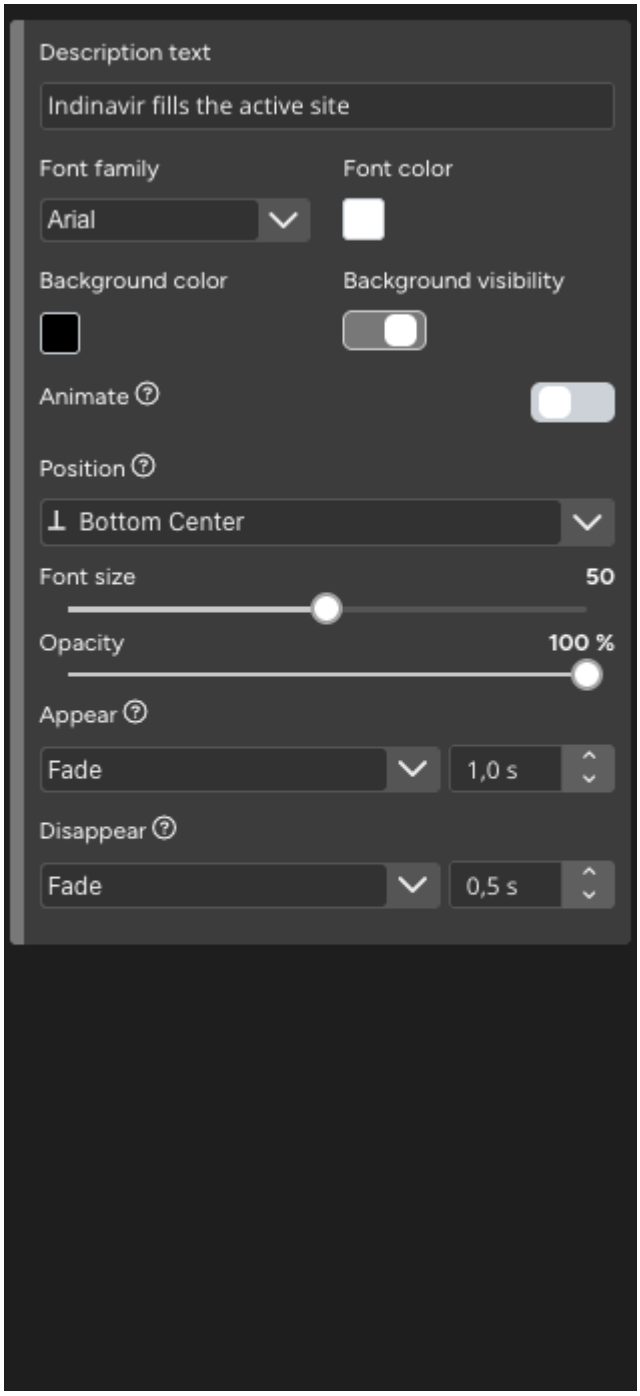
- at the **clips' speed**, or staying on one frame when both clips meet at the same frame;
- **slower** or **faster** than the clips, when the frames between them do not match the transition's length. Buttons then offer to fix it, such as *Start Clip 2 at frame 120* or *End Clip 1 at frame 95*;
- **starting again**, when the second clip starts at an earlier frame than the one the first ends on: the transition then cross-fades the two frames instead of playing.

Splitting and templates

- **Splitting** a clip that plays its trajectory keeps it going: the second part starts at the frame the first part reached.

- A **template** made from a project with a trajectory asks for a structure and a trajectory too. If the new trajectory has a different number of frames, the clips' ranges are fitted to it.

Descriptions



A description's settings in the right panel

A **description** is a text over the movie: a title, a sentence that explains what the clip shows. Descriptions have their own row in the timeline, **Descriptions**, with their own times: usually those of a clip, but they can start and end anywhere, and several can follow one another during one clip. They are part of the exported video.

Adding one

+ New > Description adds the first description at the start of the movie, 1 second long. After that, select a description (click one of its keyframes) and choose:

- **Description After** or **Description Before**: a new one next to it;

- **Cloned Description After** or **Before**: a copy of it, with its text and style, as long as it.

Drag its keyframes to set when it starts and ends, as for a clip (see The timeline). **Split** and **Join** in the timeline's toolbar also work on descriptions.

Its settings

Click one of a description's keyframes: the right panel shows its settings.

Setting	What it does
Description text	The text. Short reads best: a title under 8 words, a sentence under 12.
Font family	*Figtree*, *Arial*, *Anton*, *Caveat*, *Merriweather* or *Quicksand*.
Font color	The colour of the text.
Background color	The colour of the box behind the text.
Background visibility	Shows the box, or leaves the text alone over the stage.
Position	Where it sits on the stage: top, middle or bottom, each left, centre or right. You can also drag it on the stage; the list then says *moved*, and the button next to it puts it back where the list says.
Font size	From 20 to 80.
Opacity	From 0 (invisible) to 100.
Appear	How it comes in from its first keyframe, and for how long: *None* (at once), *Fade*, *Slide* from any side, *Grow*, *Typewriter* (letter by letter) or *Highlight* (its background sweeps in behind the text; it needs the background on).
Disappear	How it goes until its last keyframe, and for how long: *None*, *Fade*, *Slide* to any side, or *Shrink*.

Moving along its time



At its first keyframe: big, in the middle



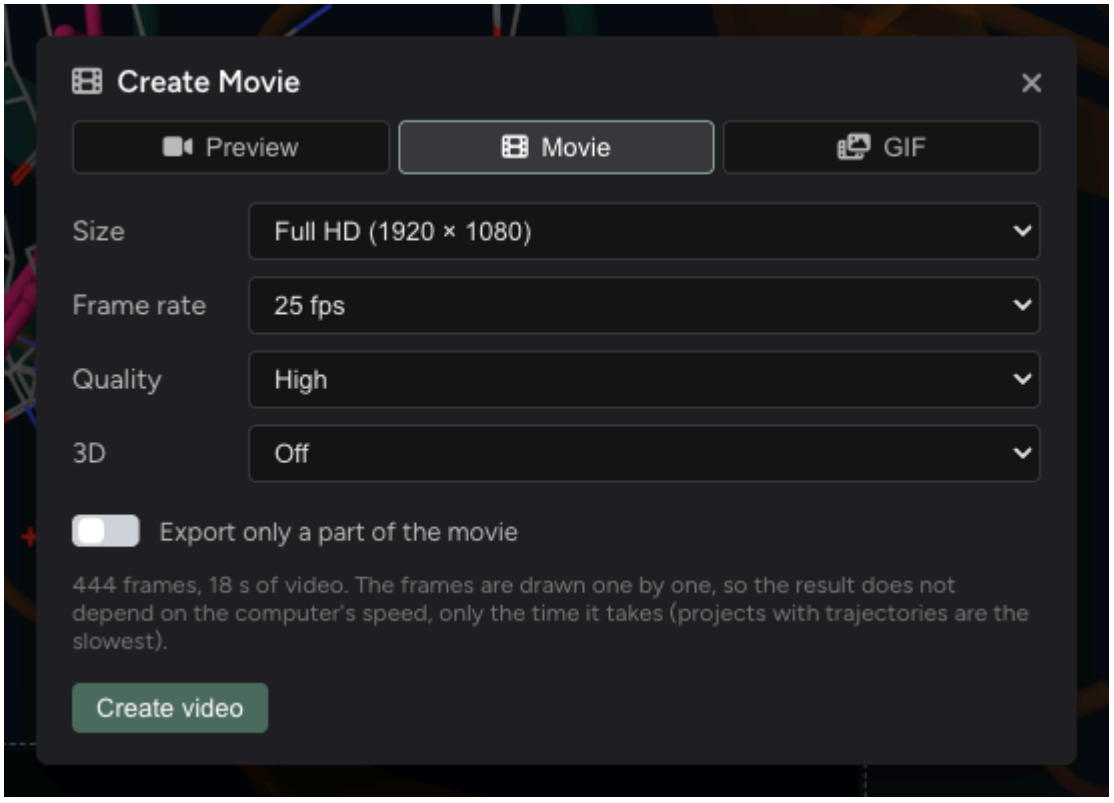
At its last keyframe: smaller, at the top

With **Animate** on, a description can change while it is on screen: its **position**, **size** and **opacity** are set at its **first keyframe** and at its **last**, and it goes from one to the other along its time. For example, a title that starts big in the middle of the stage and rises to the top as it shrinks.

- The **First keyframe** and **Last keyframe** buttons move the cursor to each end; the settings you change then are that end's. Between the two, the panel asks you to go to one of them.
- **Easing** sets how the change speeds up and slows down: *Linear*, *Ease in*, *Ease out*, or *Ease in and out*.
- Dragging the text on the stage moves it at the keyframe the cursor is on, and the button next to *Position* undoes that drag at that end only.

With **Animate** off, the description stays the same from its first keyframe to its last.

Exporting videos



The export window

The **Create Video** menu makes a video of the movie, frame by frame, at the size you choose: the result does not depend on your screen or on the size of the window, and nothing is recorded from the screen.

Menu entry	Shortcut	Starts with
New Preview	⌘P, Ctrl+P	A quick MP4: 720p, medium quality.
New Movie	⌘M, Ctrl+M	The final MP4: Full HD, high quality.
New GIF	⌘G, Ctrl+G	An animated GIF: 800 × 450, 12.5 frames per second.

All three open the same **export window**, each with its own starting settings; the tabs at its top switch between them, and each remembers what you last chose.

The settings

Setting	Choices
Size	MP4: 720p (1280 × 720), Full HD (1920 × 1080), QHD (2560 × 1440) or 4K (3840 × 2160). GIF: from 480 × 270 to 1024 × 576. A size this computer cannot encode is marked *not available here*.
Frame rate	MP4: 25 or 12.5 frames per second. GIF: 12.5, 8.3 or 5.
Quality	MP4 only: *Medium*, *High* or *Very high* (bigger files).
3D	MP4 only: a stereoscopic video (see 3D videos).
Export only a part of the movie	Off: the whole movie. On: see A part of the movie.

Videos are 16:9. The video shows the same height as the stage; only the sides depend on the stage's shape. Turn on the **Video frame** tool on the stage (see The stage) while you set the cameras: it shows the sides a video leaves out.

Every frame carries the descriptions, and a small, faint biomovies logo in the bottom right corner.

Making it

Create video (or **Create GIF**) starts. The window shows the progress, and **Cancel** stops it. A movie without a trajectory goes fast; with a trajectory, each frame takes longer, as the structure is rebuilt at each frame.

When it is done, the window plays the result and says its size, frame rate and length. Nothing is written to disk until you click **Save…**; then **Show in folder** finds it. **New export** goes back to the settings, to make another one. Closing the window before saving asks first.

A part of the movie

With **Export only a part of the movie** on, the window shows the movie's clips in small, with two handles for where the video starts and ends:

- drag a handle: it snaps to the edges of the clips, transitions and descriptions near it (hold Alt to put it anywhere), and the stage shows that moment;
- click a block to take just that block, Shift+click to extend the part to another;
- type a clip number or a time in seconds in **From** and **To**;
- zoom with the wheel or a pinch, and double-click an empty part to see the whole movie again.

The part chosen is kept while the switch is off, until the project is closed.

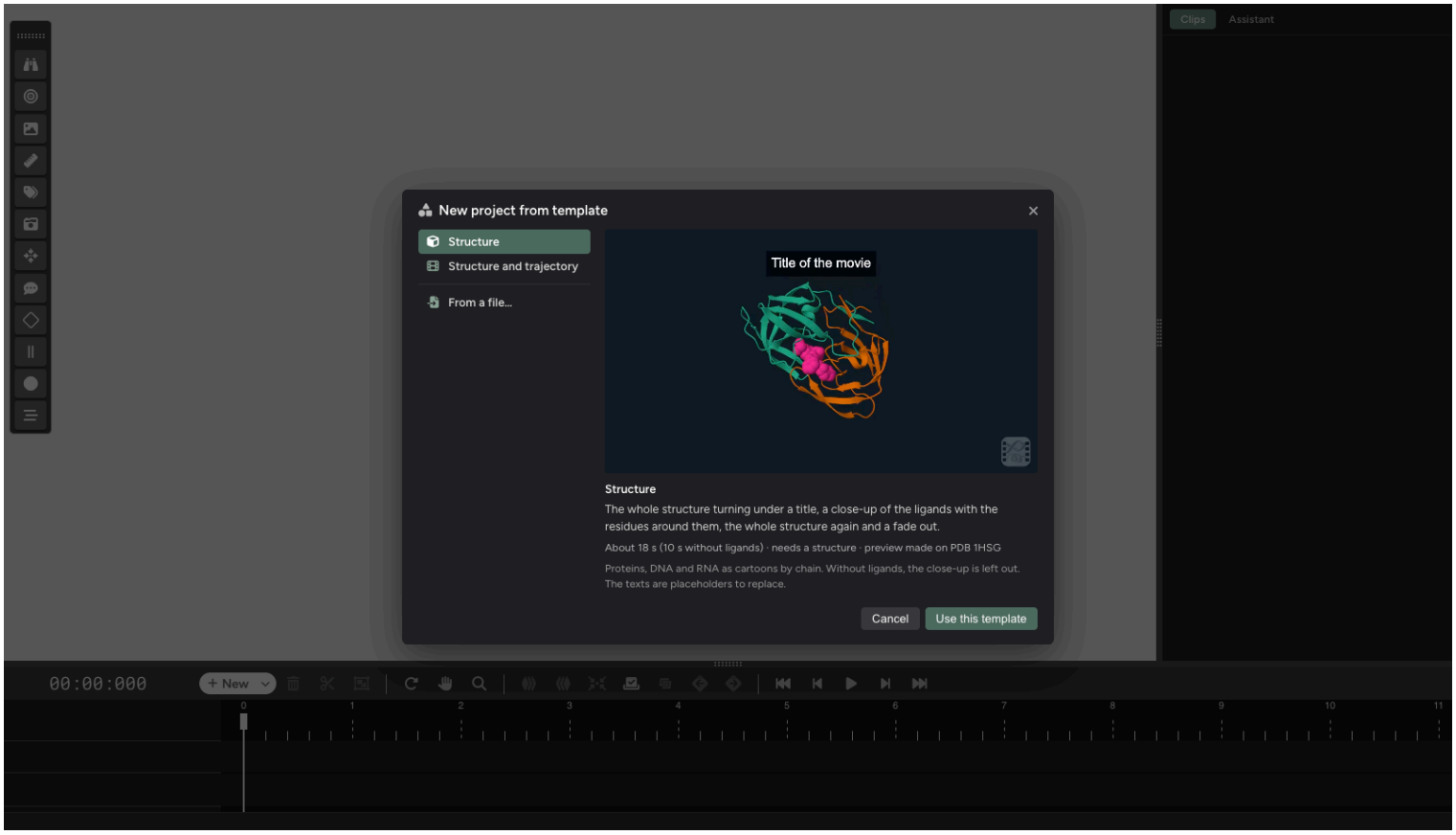
3D videos

3D makes a stereoscopic video, with one picture for each eye:

Layout	For
Side by side, half width	3D TVs and VR headsets: set the TV or the player to side-by-side 3D.
Side by side, full width	Players that take each eye at full size: the frame is twice as wide, and it takes about twice as long to make.
Top and bottom, half height	TVs and players set to top-and-bottom 3D.
Anaglyph	Red / cyan glasses, on any screen; the colours change a little.

Depth sets how strong the 3D is, and **Screen plane** how far the screen sits in the scene: what is in front of it comes out of the screen, what is behind goes in. The descriptions stay on the screen plane. A still of the movie in 3D shows the effect before you start.

Templates



The templates window

A **template** is a movie without its structure: its clips and their times, representations and cameras, transitions, fades, descriptions, labels and measurements, and its trajectory settings. Applied to another structure, it makes the same movie of it. A template file (**.tbiomovies**) is small, as it holds no structure and no trajectory: an easy way to share how a movie is made.

Starting from a template

New project from template on the start screen (⇧⌘I on macOS, Ctrl+Alt+I on Windows and Linux; also in the *Project* menu) opens the **templates window**:

- **Ready-made templates**, each with a preview video made on an example structure, what it shows, how long it is and notes on what to expect:
  - **Structure**: the whole structure turning under a title, a close-up of the ligands with the residues around them, the whole structure again, and a fade out (about 18 s; 10 s for a structure without ligands, where the close-up is left out).
  - **Structure and trajectory**: a still overview under a title, the whole trajectory played while the camera turns, an ending played backwards, and a fade out (about 13 s).
- **From a file…**: a **.tbiomovies** file of your own. The window shows what it holds (its length, clips, transitions, descriptions, labels…) and what may not fit another structure (see below).

After **Use this template**, the start screen asks for the structure to use with it:

- for a structure template, a **structure file** or a **structure code** (PDB, AlphaFold DB, UniProt);
- for a trajectory template, the **structure and trajectory files** or an **MDposit** simulation.

Back returns to the start screen without a template. The texts of the ready-made templates are placeholders: change them in the editor (see Descriptions).

Applying one to the open project

Project > Apply Template… opens the same window from the editor, with the templates that fit the open project (with or without a trajectory). **Apply** makes a new project from the open one's structure (and trajectory, and whether it shows the biological assembly) with the template. The open project is closed first, so biomovies asks to save it.

Making your own

Project > Export Project As Template saves the open project as a **.tbiomovies** file: everything but the structure and the trajectory.

What fits any structure and what may not:

- **Selections by keyword** (*Protein*, *Ligand*, `not water` …) fit any structure.
- **Selections naming residues, chains or atoms** (`25:A` , `:B`) only work where those exist; so do **labels** and **measurements**, which are on atoms. The templates window lists them before you use the template.
- A **trajectory** with a different number of frames: the clips' ranges are fitted to it.

AI assistants

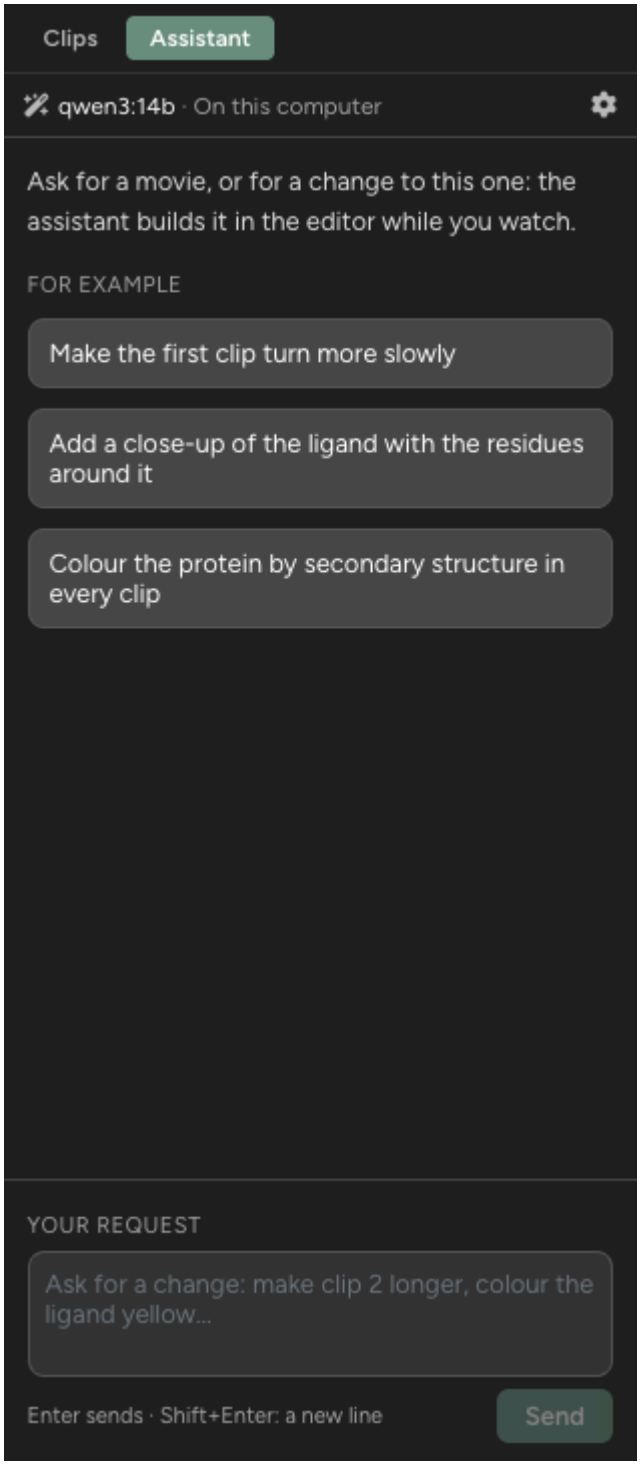
An **AI assistant** can build and change movies in biomovies for you: ask it (*"make a 20-second movie of 1HSG showing how the inhibitor sits in the protease"*) and watch the movie being made in the editor. It opens the structure, plans the clips, builds them, looks at the result and corrects it.

There are two ways:

- **The assistant in biomovies**: a panel in the editor, with a language model from a provider you choose (Google, Anthropic, OpenRouter, a model on your own computer…). See The assistant in biomovies.
- **Assistants in other apps**, connected to biomovies: see Assistants in other apps.

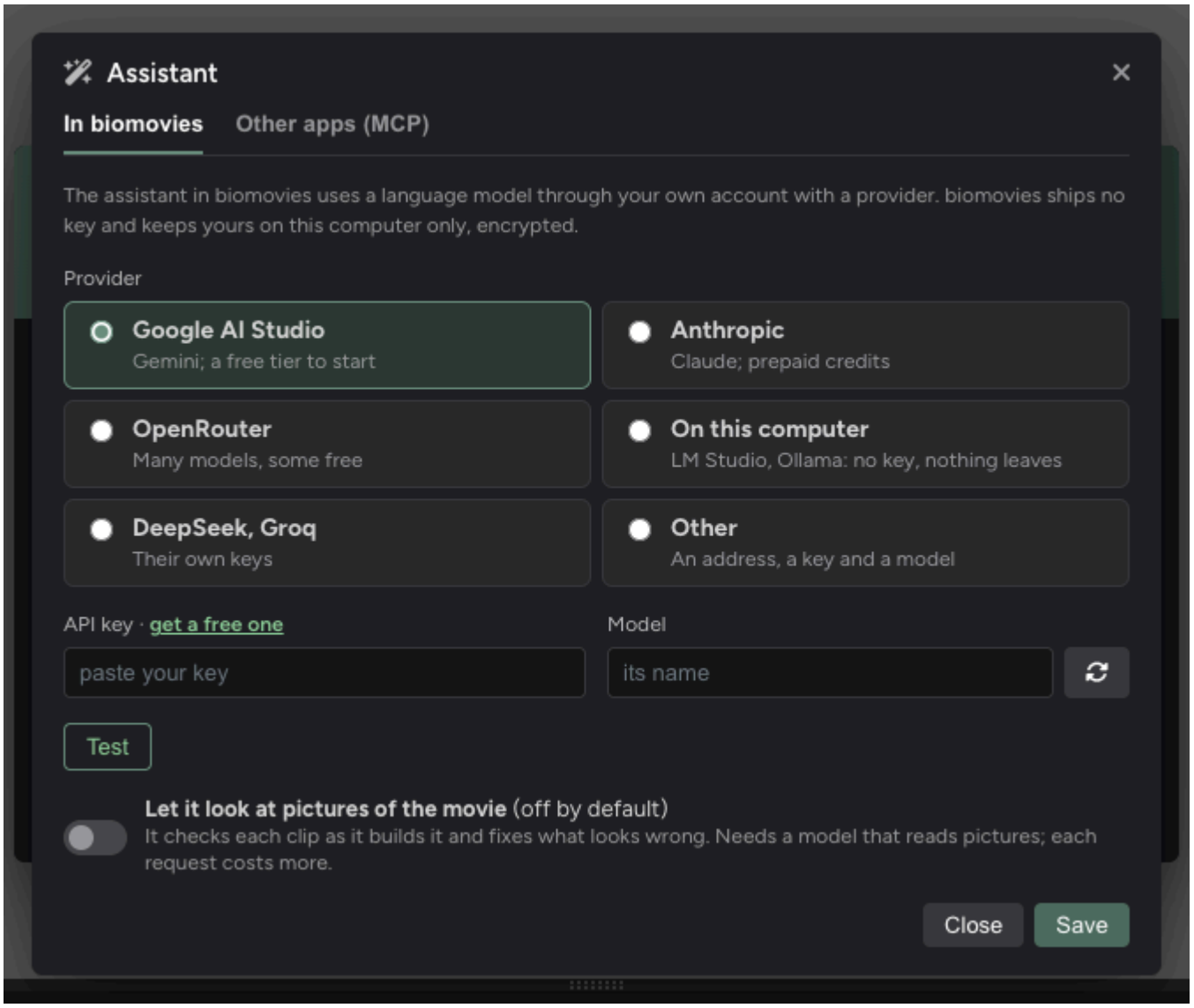
biomovies includes no AI and no account of its own: you use your own key or account, and the provider's plan applies.

The assistant in biomovies



The assistant's panel, in the editor's right column

Setting it up



Assistant > Connection..., In biomovies

Assistant > Connection..., tab **In biomovies**:

1. Choose a **provider**:

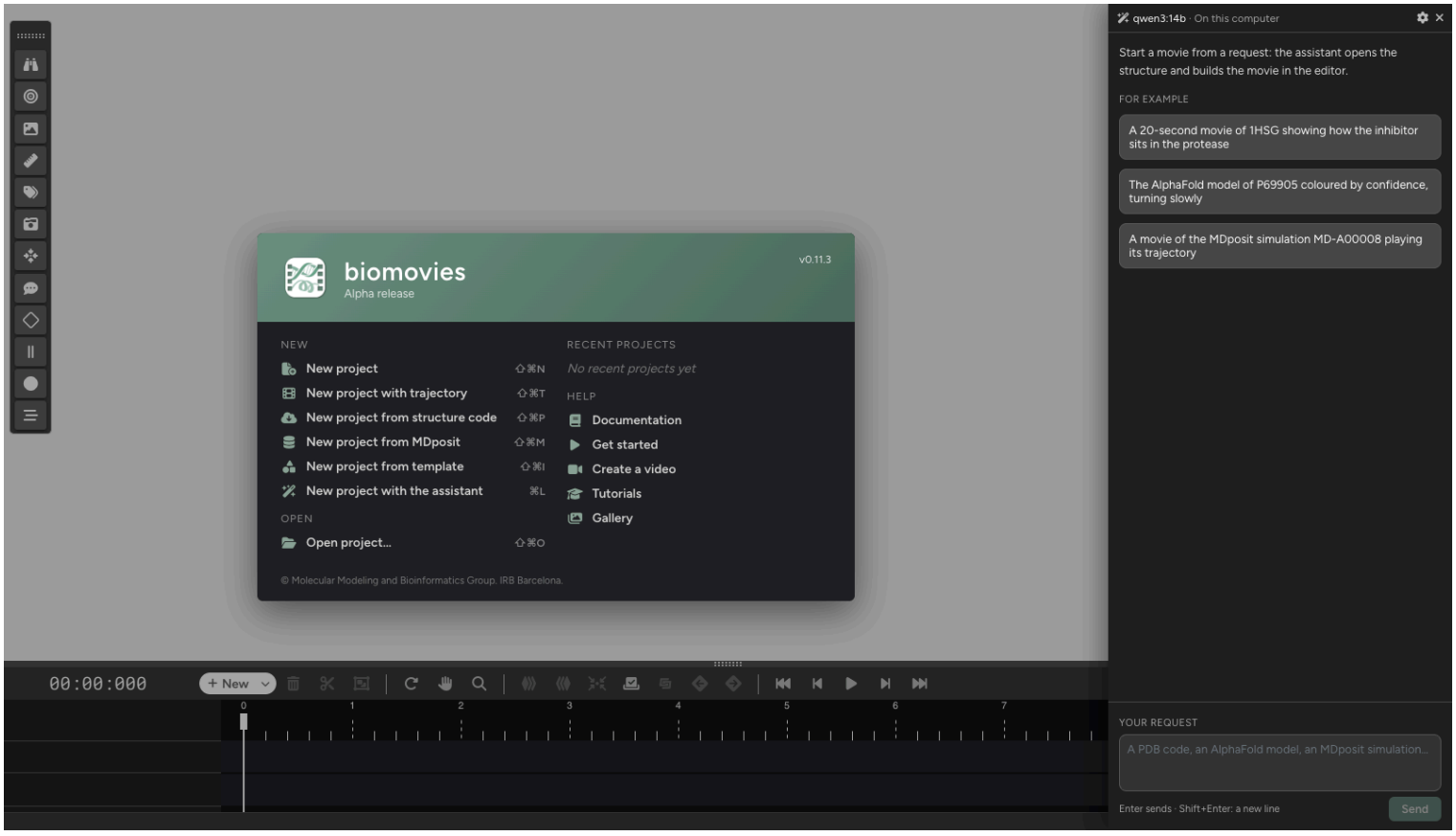
Provider	What to know
Google AI Studio	Gemini models, with a free tier to start (a key at aistudio.google.com/apikey). The newest models are often overloaded; an older *Flash* one usually answers.
Anthropic	Claude models, with prepaid credits in Anthropic's console. Claude Opus is the most capable; Sonnet and Haiku cost less.
OpenRouter	Many models from many companies under one key. Choose a large, recent model that takes tools; the free ones (`:free`) allow only a few requests a day.
On this computer	A model running in LM Studio or Ollama: no key, nothing leaves the computer. On a laptop a request takes minutes; give the model a context length of at least 32k.
DeepSeek, Groq	Their own keys. Groq's free tier allows too few tokens a minute for the assistant.
Other	Any server that speaks the OpenAI chat API: its address, a key if it needs one, and a model.

1. Paste your **API key** (the link next to it says where to get one). biomovies keeps it on this computer only, encrypted by the system; it is never in a project and never sent anywhere but to the provider.
2. Choose a **model**. The list shows only the models that can chat and use tools; the refresh button asks the provider for its list again.
3. **Test**: biomovies asks the model a short question and says whether it answered, whether it uses tools (the assistant needs them) and whether it reads pictures.
4. **Let it look at pictures of the movie** (off by default): the assistant then checks the pictures of what it built and fixes what looks wrong. It needs a model that reads pictures, and each request costs more.
5. **Save**.

On **macOS**, the first time biomovies uses the saved key it asks to use its keychain entry, *"biomovies Safe Storage"*: type your Mac's password and choose **Always Allow**. After an update of biomovies, macOS may ask once more.

What it costs. Each step of the assistant sends the model about 18,000 tokens (the tools and the guide), and a movie takes 10 to 30 steps. A free tier with a small limit per minute may refuse it (the message says so). With Claude, a movie costs from cents (Haiku) to a dollar or two (Opus), roughly. The panel says what each request used.

The panel



New project with the assistant, on the start screen

- **Open it** with the **Assistant** tab of the editor's right column (beside **Clips**), with **Assistant > Assistant Panel** (⌘L on macOS, Ctrl+L on Windows and Linux), or with **New project with the assistant** on the start screen. Without a model or a key, the panel offers **Set up the assistant**.

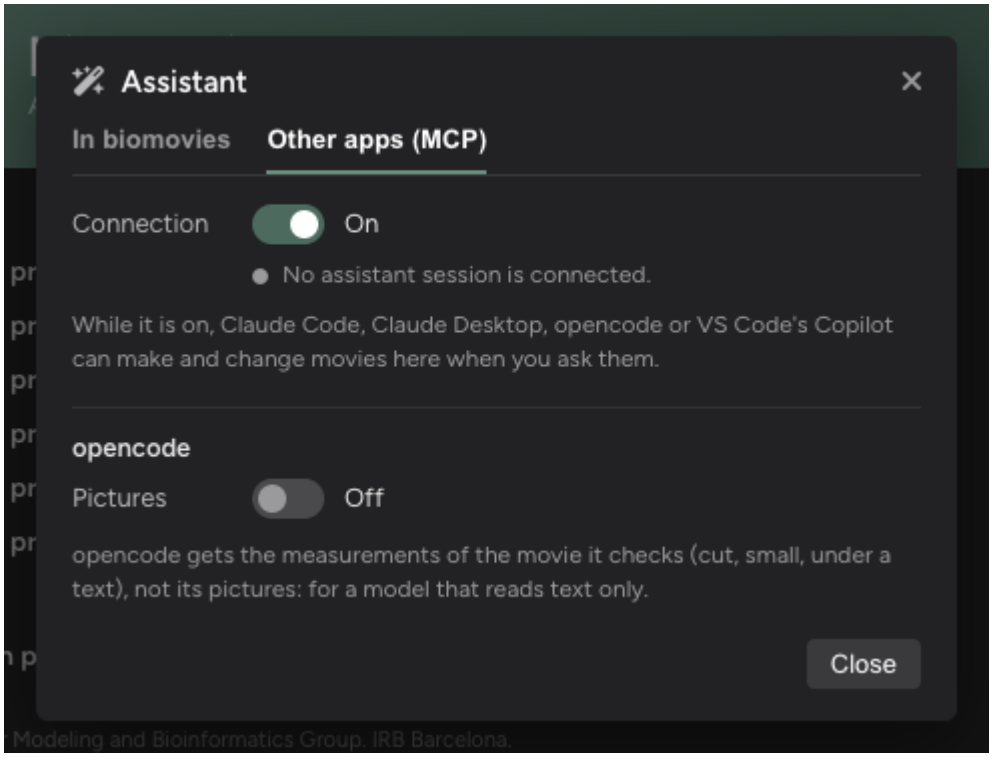
- **Ask**: type your request and press Enter (Shift+Enter for a new line), or click one of the examples. On the start screen, the request creates the project; in the editor, it changes the movie open.
- **Before your first request** (once each time biomovies opens), a window says what goes to the provider (what you write, a summary of the structure, the movie's outline, its pictures if *Pictures* is on) and what does not (the structure and trajectory files, the project file). **Continue** sends the request, **Not now** keeps it in the box; tick **Do not show this again** to skip it from then on.
- **While it works**, the editor is locked and the panel shows each step as it happens. **Stop**, or Esc, stops it.
- **When it is done**, it says what it made. Under its answer:
  - **N steps**: click to see what it ran, and what failed;
  - **Undo all of it**: undoes the whole request in one go, while nothing has been changed since (after that, use **Edit > Undo** step by step);
  - **Preview the movie**: plays it from the start;
  - how many calls to the model the request took, and about how many tokens.
- The conversation lasts while biomovies is open (it is not saved in the project): the assistant remembers your earlier requests. The broom button starts a new conversation; the gear opens the settings.

Assistants in other apps

biomovies works with four assistants of other apps, which you use with your own account:

- **Claude Code** (Anthropic's assistant in the terminal, or in VS Code);
- **Claude Desktop** (its chats, Cowork and its Code tab);
- **opencode** (with any model it can reach, including models running on your own computer or your institution's servers);
- **VS Code's Copilot** (its chat in Agent mode).

Connecting an assistant



Assistant > Connection..., Other apps (MCP)

Assistant > Connection…, tab **Other apps (MCP)**, on the start screen or in the editor. (MCP, the Model Context Protocol, is the standard these apps use to reach biomovies' tools.)

1. Turn the **Connection** on. While it is on, the assistants can work on the open project (or start one from the start screen) when you ask them. biomovies remembers it, also after a restart, until you turn it off.
2. For each assistant installed on this computer that does not have the biomovies tools yet, the window offers a button: **Add to Claude Code**, **Add to Claude Desktop**, **Add to opencode** or **Add to VS Code**. Done once per assistant.
3. Start a new session of the assistant (Claude Code, opencode), reopen Claude Desktop, or open VS Code's chat in Agent mode: biomovies' tools are now among its tools.

A few details:

- **Claude Desktop** must be closed while you add biomovies to it: it keeps its settings in the same file.
- **opencode** and **Claude Desktop** settings are edited only if they are plain files biomovies can read; a copy of the file as it was stays next to it (*.biomovies-backup*).
- **VS Code** asks you to trust the biomovies server the first time it starts it.
- When adding fails, the window shows the lines to paste (or the command to run) instead, with a **Copy** button.
- When biomovies is moved or updated, the window may say an assistant's tools *run a copy of biomovies that is no longer there*: click **Update** next to it.

Asking for a movie

Ask in plain words, in the assistant's chat:

- *"Make a movie of PDB 1HSG: the whole protease, then a close-up of the inhibitor."*
- *"The AlphaFold model of P69905 coloured by confidence, turning slowly."*
- *"Make clip 2 longer and colour the ligand yellow."* (on the open project)

Claude Code, opencode and Copilot also offer two ready requests:

Request	Claude Code, opencode	VS Code's Copilot	What happens
movie	`/biomovies:movie 1HSG`	`/mcp.biomovies.movie`	The assistant proposes a plan for a movie of that structure and builds it when you agree.
paper	`/biomovies:paper` followed by the PDF's path	`/mcp.biomovies.paper`	It reads the paper, finds its structure, proposes a storyboard with the paper's own words and builds it when you agree.

In Claude Desktop, ask in plain words: its chats do not list these requests.

A few things to know:

- **New movies** start from the start screen: if a project is open, the assistant asks you to save and close it first.
- For a **PDB entry** whose biological assembly differs from the structure as deposited, the assistant asks which one you want.
- A movie shows **one structure**: comparing two (with and without a drug) is not possible yet.

While it works

While the assistant changes the movie, a card at the top says what it is doing, and the editor is locked so that nothing changes under it; you watch the movie being built. To stop it, stop the assistant in its own window (Esc in Claude Code); if the editor stays locked, the card says when it unlocks, or press **Esc** on it.

When it is done:

- **Edit > Undo** (⌘Z, Ctrl+Z) undoes the **whole request** in one step.
- Everything it made can be changed in the editor as usual.
- The assistant tells what it made, clip by clip, what it checked and what it corrected.

Pictures

To check its work, the assistant looks at pictures of the movie's key moments, with measurements of how its subject sits in each one (cut at an edge, too small, under a text).

- **Claude Code**, **Claude Desktop** and **Copilot** get the pictures.
- **opencode** gets the measurements only, as many of its models read text only. With a model that reads images, turn **Pictures** on under opencode in **Assistant > Connection…**, tab **Other apps (MCP)**.

Scripts

What an assistant does, in biomovies or in another app, is a list of the editor's commands. biomovies can keep that list as a **script**, a small .json file, and run it again later.

Saving one

Assistant > Save the Assistant's Commands as Script… saves the commands the assistants ran on the project you are working on, in order, with your requests as its description: from the one that created it, or from the first one they ran after you opened it. One project per script, so that it runs again: what they did to a project you closed before is not in it (save that one before you close it). Only the commands that worked are kept, and those of a request you undid afterwards too: save right after the requests you want to keep.

Running one

Assistant > Run Script… asks for the .json file and runs it:

- **On the start screen**: a script that starts by creating the project (all those saved from a new movie do) makes the project and builds the movie in it.
- **In the editor**: the script changes the open project. Everything it does is **one undo step** (**Edit > Undo**).

It runs the commands one after the other and **stops at the first one that fails**, saying which and why; what ran before it stays (and is undone in one step). Next to the script, biomovies writes what each command gave (*<script>.results.json*) and the pictures some commands take (*<script>.screenshot-<n>.png*).

What for

- **The same movie again**, on another computer or after a change in biomovies, without asking an assistant (and without paying for it again).
- **Sharing how a movie was made**: send the script with the paper or to a colleague.
- **A starting point**: run it on the start screen and change the movie in the editor, as with a template (for a template of your own that fits any structure, see Templates).

Tips and shortcuts

Making good movies

- **From the whole to the detail**: start with the whole structure, then fly to what the movie is about (a pocket, a ligand, a mutation, an interface) with a transition.
- **One idea per clip**, 3 to 6 seconds each; transitions of 1 to 2 seconds. **Cloned Clip After** is the usual way to go on: change only what differs, and a transition animates the change.
- **Keep the context, faded**: the rest of the protein at an opacity of 20 to 40 while the subject is solid, in a colour of its own (see Representations).
- **Gentle cameras**: a turn of 30 to 90 degrees during a clip gives the sense of 3D; more than about 120 is too fast.
- **Short texts**: a title under 8 words, a description under 12, one at a time.
- **A dark background** makes the colours stand out; a light one suits print.
- **End with a fade out** of about a second.
- Turn on the **Video frame** tool before setting the cameras, so that nothing important falls outside the video.

Working with biomovies

- **Save often**: biomovies is in its alpha version. The status bar says when there are unsaved changes, and closing a project with changes asks first.
- **Undo** (⌘Z, Ctrl+Z) undoes any change, a whole assistant's request included.
- **Look around** without changing any keyframe with **Explore** (see The stage).
- **Open projects** from the start screen's **Recent projects**, or double-click a **.biomovies** or **.tbiomovies** file.
- **Share** a movie as a **.biomovies** file (with the structure and the trajectory inside), or as a template (**.tbiomovies**: small, without them).

Big projects

A project with hundreds of representations, or a structure of millions of atoms in many clips, would not fit in the computer's memory if every clip were kept on the stage. biomovies then switches to the **memory saving mode** by itself, and a message says so: only the clips, transitions and fades near the cursor are kept on the stage, and the others are prepared as the cursor gets near them. Playing waits for them (*Preparing…* on the stage); the exported videos are the same.

Frequent questions

I drag on the stage and the camera does not change. The mouse moves a clip's camera only with **one clip keyframe selected** in the timeline (click it). With the cursor between keyframes, the stage shows the movie but keeps its camera; to look around anyway, use **Explore**.

+ New > Transition says something is missing. Select the last keyframe of a clip and Shift+click the first keyframe of the next one, with at least 0.4 s between them (drag the second clip to the right to make room).

A selection matches no atoms. Residue numbers and chain names are the file's own, which may differ from a paper's. The **Sequences** tool shows the chains and their numbering.

pLDDT (confidence) is greyed out. The structure is experimental, or its file has no confidence scores: the reason is under the scheme's name in the list.

The video shows less than the stage at the sides. Videos are 16:9: turn on the **Video frame** tool to see what they leave out.

My computer warns about biomovies when installing it. See Installation: biomovies is not signed with a paid certificate, so the system asks you to confirm the first time.

Something went wrong. Undo, save the project under a new name, and tell us with **Help > Report Issue**.

Licences

biomovies is free software, under the **GNU General Public License v3.0**: **Help > License** shows it, and **Help > Third-Party Licenses** the licences of everything it is built on.

Shortcuts

The menus show each entry's shortcut. Some entries work only on the start screen (*start screen* below) and the rest in the editor.

The biomovies menu

Menu entry	macOS	Windows	Linux
Hide biomovies	⌘H		
Hide Others	⌥⌘H		
Quit biomovies	⌘Q		
Exit		Ctrl+Q	Ctrl+Q

Edit

Menu entry	macOS	Windows	Linux
Undo	⌘Z	Ctrl+Z	Ctrl+Z
Redo	⇧⌘Z	Ctrl+Shift+Z	Ctrl+Shift+Z

Project

Menu entry	macOS	Windows	Linux
Save Project	⌘S	Ctrl+S	Ctrl+S
Save Project As	⇧⌘S	Ctrl+Shift+S	Ctrl+Shift+S
Load Trajectory…	⇧⌘L	Ctrl+Shift+L	Ctrl+Shift+L
Export Project As Template	⇧⌘E	Ctrl+Shift+E	Ctrl+Shift+E
Close Project	⌘W	Ctrl+W	Ctrl+W
New Project (start screen)	⇧⌘N	Ctrl+Shift+N	Ctrl+Shift+N
New Project from Structure Code… (start screen)	⇧⌘P	Ctrl+Shift+P	Ctrl+Shift+P
New Project from MDposit… (start screen)	⇧⌘M	Ctrl+Shift+M	Ctrl+Shift+M
New Project with Trajectory (start screen)	⇧⌘T	Ctrl+Shift+T	Ctrl+Shift+T
Open Project (start screen)	⇧⌘O	Ctrl+Shift+O	Ctrl+Shift+O
New Project from Template… (start screen)	⇧⌘I	Ctrl+Alt+I	Ctrl+Alt+I

View

Menu entry	macOS	Windows	Linux
Stage	⌥⇧S	Alt+Shift+S	Alt+Shift+S
Timeline	⌥⇧T	Alt+Shift+T	Alt+Shift+T
Representations	⌥⇧R	Alt+Shift+R	Alt+Shift+R
Tools	⌥⇧O	Alt+Shift+O	Alt+Shift+O
Status Bar	⌥⇧B	Alt+Shift+B	Alt+Shift+B
Explore Stage	⌥⇧E	Alt+Shift+E	Alt+Shift+E
Light Theme	⌥⇧H	Alt+Shift+H	Alt+Shift+H

Timeline

Menu entry	macOS	Windows	Linux
New Item > Create New Clip	⇧⌘C	Ctrl+Shift+C	Ctrl+Shift+C
New Item > Create New Cloned Clip	⌥⌘C	Ctrl+Alt+C	Ctrl+Alt+C
New Item > Create New Transition	⇧⌘T	Ctrl+Shift+T	Ctrl+Shift+T
New Item > Create New Fade In Transition	⇧⌘I	Ctrl+Alt+I	Ctrl+Alt+I
New Item > Create New Fade Out Transition	⇧⌘O	Ctrl+Shift+O	Ctrl+Shift+O
New Item > Create New Description	⇧⌘D	Ctrl+Shift+D	Ctrl+Shift+D
Split Clip or Description at Cursor	⇧⌘X	Ctrl+Shift+X	Ctrl+Shift+X
Join Clips or Descriptions	⇧⌘J	Ctrl+Shift+J	Ctrl+Shift+J
Controls > Reset Timeline	⌥⌘R	Ctrl+Alt+R	Ctrl+Alt+R
Controls > Pan Mode	⌥⌘A	Ctrl+Alt+A	Ctrl+Alt+A
Controls > Zoom Mode	⌥⌘Z	Ctrl+Alt+Z	Ctrl+Alt+Z
Keyframes Actions > Align All to First	⌥⇧⌘↑	Ctrl+Alt+Shift+↑	Ctrl+↑
Keyframes Actions > Align All to Cursor Keyframe	⌥⇧⌘C	Ctrl+Alt+Shift+C	Ctrl+Alt+Shift+C
Keyframes Actions > Align All to Last	⌥⇧⌘↓	Ctrl+Alt+Shift+↓	Ctrl+↓
Keyframes Actions > Select all Keyframes	⇧⌘K	Ctrl+Shift+K	Ctrl+Shift+K
Keyframes Actions > Associate Transition to Clip	⇧⌘A	Ctrl+Shift+A	Ctrl+Shift+A
Keyframes Actions > Move to Left	⇧⌘←	Ctrl+Shift+←	Ctrl+Shift+←
Keyframes Actions > Move to Right	⇧⌘→	Ctrl+Shift+→	Ctrl+Shift+→
Player > Go to First Frame	⇧↑	Shift+↑	Shift+↑
Player > Go to Previous Frame	⇧←	Shift+←	Shift+←
Player > Play / Pause	⇧P	Shift+P	Shift+P
Player > Go to Next Frame	⇧→	Shift+→	Shift+→
Player > Go to Last Frame	⇧↓	Shift+↓	Shift+↓

Create Video

Menu entry	macOS	Windows	Linux
New Preview	⌘P	Ctrl+P	Ctrl+P
New Movie	⌘M	Ctrl+M	Ctrl+M
New GIF	⌘G	Ctrl+G	Ctrl+G

Assistant

Menu entry	macOS	Windows	Linux
Assistant Panel	⌘L	Ctrl+L	Ctrl+L

Help

Menu entry	macOS	Windows	Linux
Documentation	⌘D	Ctrl+D	Ctrl+D

Other keys and gestures

Where	Keys	What they do
Stage	Drag, right-drag, scroll	Turn, move, zoom (the camera of the selected clip keyframe).
Stage	Alt+click two atoms	A distance.
Stage	Alt+right-click three atoms	An angle.
Stage	Alt+Shift+click an atom	A label with the name of its residue.
Stage	Esc	Leaves **Explore**.
Timeline	Shift+click	Adds a keyframe or an item to the selection.
Timeline	Double-click a bar, or ⌘+click / Ctrl+click	Selects the whole item.
Timeline	Ctrl+scroll, or a trackpad pinch	Zooms the timeline.
Trajectory window	⌘⌥1 / ⌘⌥2 (Ctrl+Alt+1 / 2)	The range's first frame down / up.
Trajectory window	⌘⌥3 / ⌘⌥4 (Ctrl+Alt+3 / 4)	The range's last frame down / up.
Sequences	Click, drag, Shift+click	Adds or removes residues of the selected representation.
Export window	Alt while dragging a handle	Puts it anywhere, not only at an item's edge.
An assistant working	Esc	Unlocks the editor once you have stopped the assistant.

On macOS, Alt is the ⌥ **Option** key.